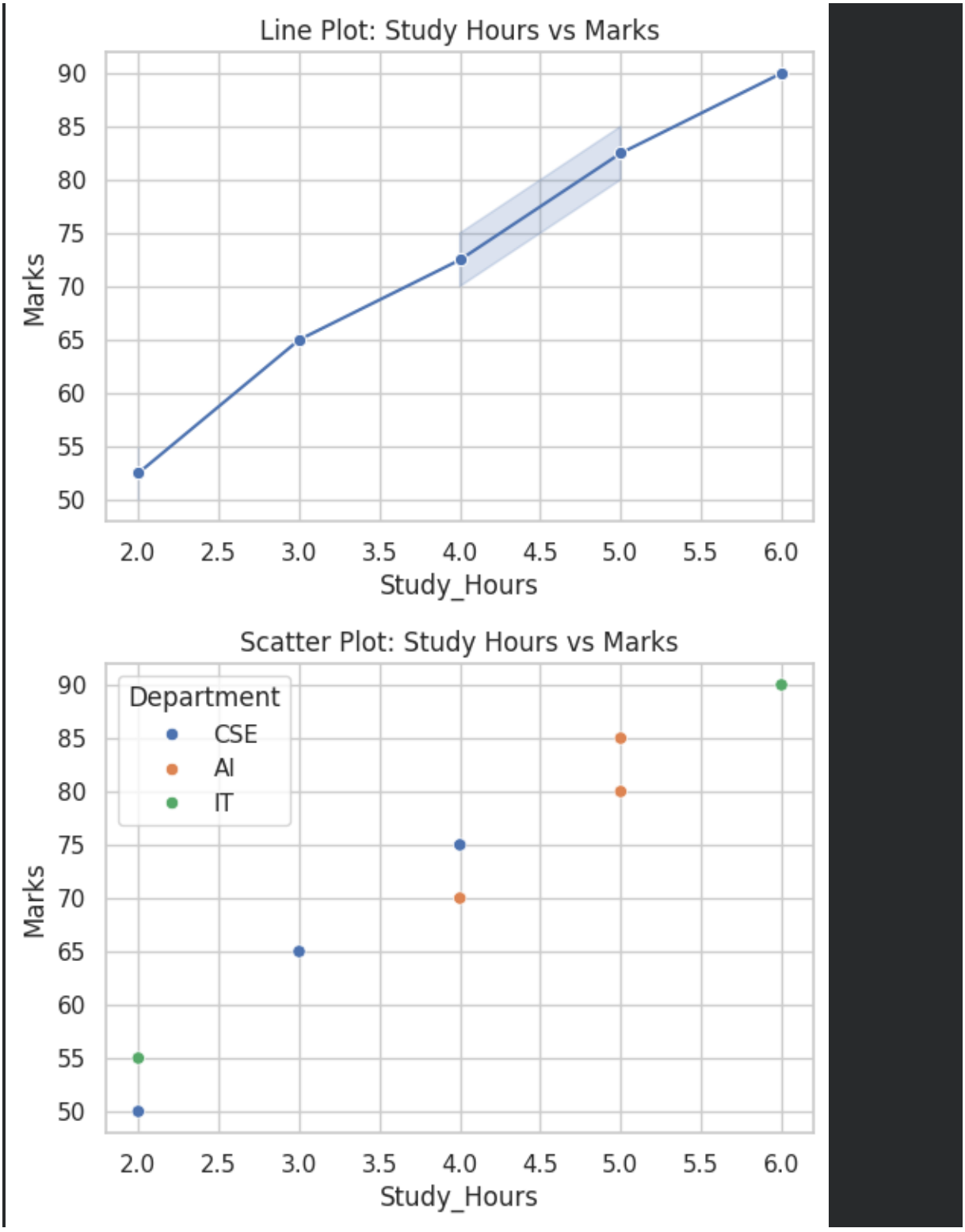
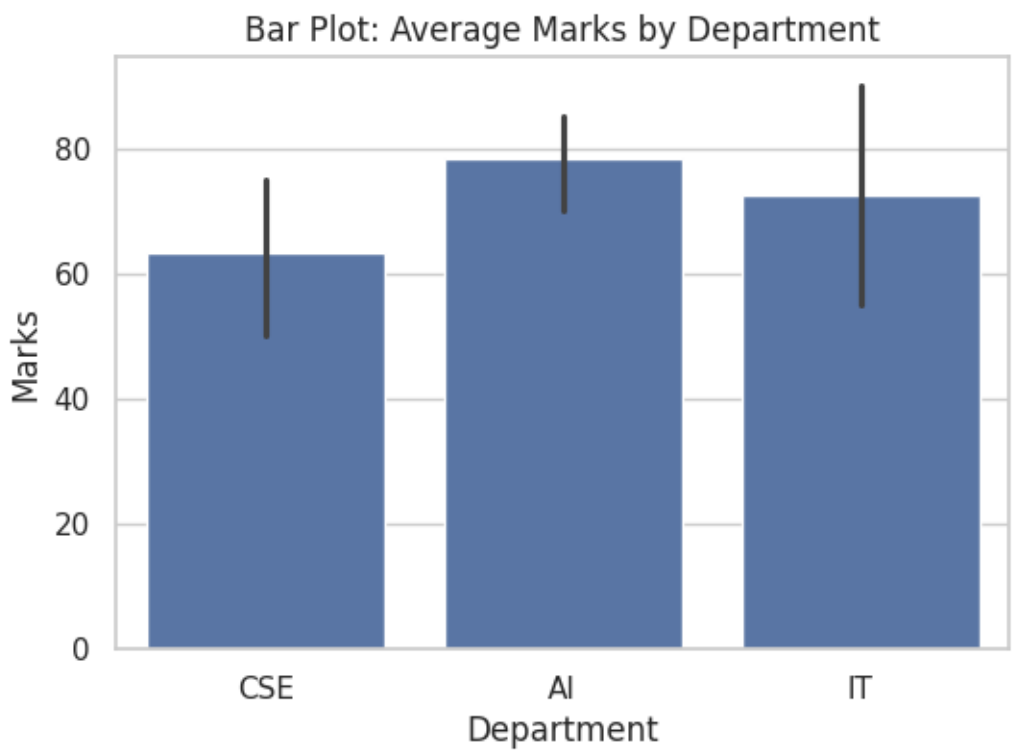
```
# Seaborn Important Graphs and Functions in One Program
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
# ----------------------------------------
# 1. Create Sample Dataset
# ----------------------------------------
data = {
   "Student": ["Arun", "Bala", "Chitra", "Divya",
               "Ezhil", "Farhan", "Geetha", "Hari"],
   "Department": ["CSE", "AI", "CSE", "AI",
                  "IT", "IT", "CSE", "AI"],
   "Gender": ["Male", "Male", "Female", "Female",
              "Male", "Male", "Female", "Male"],
   "Study_Hours": [2, 4, 3, 5, 6, 2, 4, 5],
   "Marks": [50, 70, 65, 85, 90, 55, 75, 80],
   "Attendance": [70, 85, 80, 95, 90, 65, 88, 92]
}
df = pd.DataFrame(data)
print("Sample DataFrame:")
print(df)
# ----------------------------------------
# 2. Set Seaborn Theme
# ----------------------------------------
sns.set_theme(style="whitegrid")
# ----------------------------------------
# 3. Line Plot
# ----------------------------------------
plt.figure(figsize=(6, 4))
sns.lineplot(data=df, x="Study_Hours", y="Marks", marker="o")
plt.title("Line Plot: Study Hours vs Marks")
plt.show()
# ----------------------------------------
# 4. Scatter Plot
# ----------------------------------------
plt.figure(figsize=(6, 4))
sns.scatterplot(data=df,
                x="Study_Hours",
                y="Marks",
                hue="Department")
plt.title("Scatter Plot: Study Hours vs Marks")
plt.show()
# ----------------------------------------
# 5. Bar Plot
# ----------------------------------------
plt.figure(figsize=(6, 4))
sns.barplot(data=df, x="Department", y="Marks")
plt.title("Bar Plot: Average Marks by Department")
plt.show()
# ----------------------------------------
```

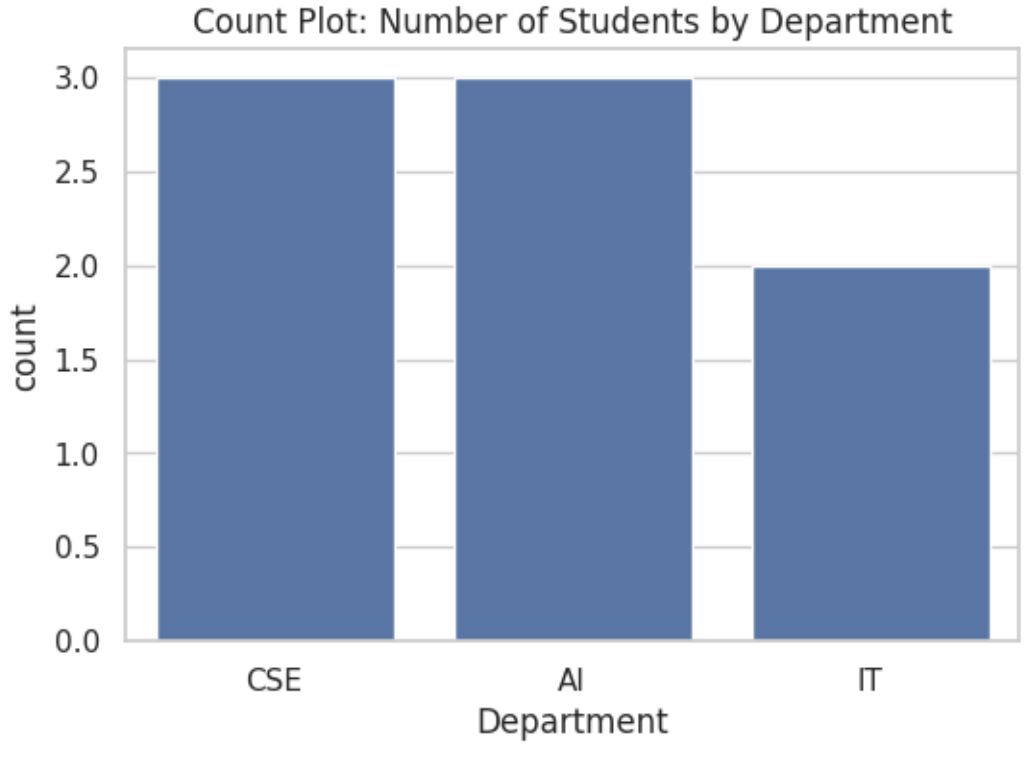
```
# 6. Count Plot
# ----------------------------------------
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Department")
plt.title("Count Plot: Number of Students by Department")
plt.show()
# ----------------------------------------
# 7. Box Plot
# ----------------------------------------
plt.figure(figsize=(6, 4))
sns.boxplot(data=df, x="Department", y="Marks")
plt.title("Box Plot: Marks Distribution by Department")
plt.show()
# ----------------------------------------
# 8. Violin Plot
# ----------------------------------------
plt.figure(figsize=(6, 4))
sns.violinplot(data=df, x="Department", y="Marks")
plt.title("Violin Plot: Marks Distribution by Department")
plt.show()
# ----------------------------------------
# 9. Histogram
# ----------------------------------------
plt.figure(figsize=(6, 4))
sns.histplot(data=df, x="Marks", bins=5, kde=True)
plt.title("Histogram: Marks Distribution")
plt.show()
# ----------------------------------------
# 10. KDE Plot
# ----------------------------------------
plt.figure(figsize=(6, 4))
sns.kdeplot(data=df, x="Marks", fill=True)
plt.title("KDE Plot: Marks Density")
plt.show()
# ----------------------------------------
# 11. Pair Plot
# ----------------------------------------
sns.pairplot(df[["Study_Hours", "Marks", "Attendance"]])
plt.suptitle("Pair Plot: Numeric Variable Relationships", y=1.02)
plt.show()
# ----------------------------------------
# 12. Heatmap
# ----------------------------------------
correlation = df[["Study_Hours", "Marks", "Attendance"]].corr()
plt.figure(figsize=(6, 4))
sns.heatmap(correlation, annot=True, cmap="coolwarm")
plt.title("Heatmap: Correlation Matrix")
plt.show()
# ----------------------------------------
# 13. Regression Plot
```

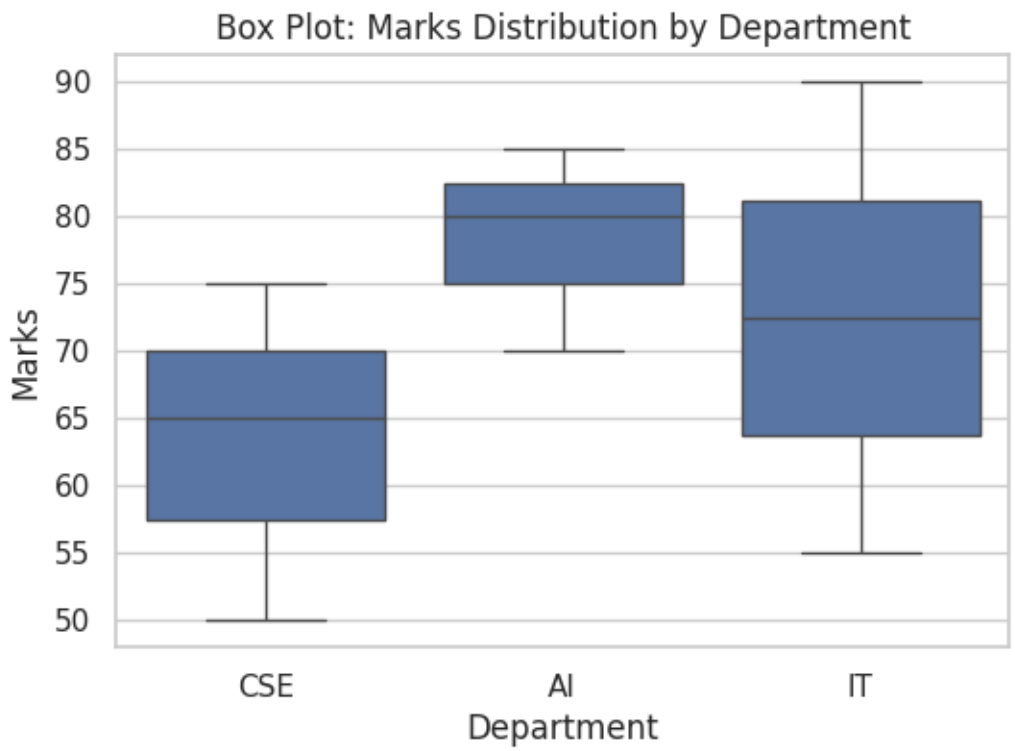
```
# ----------------------------------------
plt.figure(figsize=(6, 4))
sns.regplot(data=df, x="Study_Hours", y="Marks")
plt.title("Regression Plot: Study Hours vs Marks")
plt.show()
# ----------------------------------------
# 14. Strip Plot
# ----------------------------------------
plt.figure(figsize=(6, 4))
sns.stripplot(data=df, x="Department", y="Marks")
plt.title("Strip Plot: Marks by Department")
plt.show()
# ----------------------------------------
# 15. Swarm Plot
# ----------------------------------------
plt.figure(figsize=(6, 4))
sns.swarmplot(data=df, x="Department", y="Marks")
plt.title("Swarm Plot: Marks by Department")
plt.show()
print("Seaborn Program Completed Successfully")
```

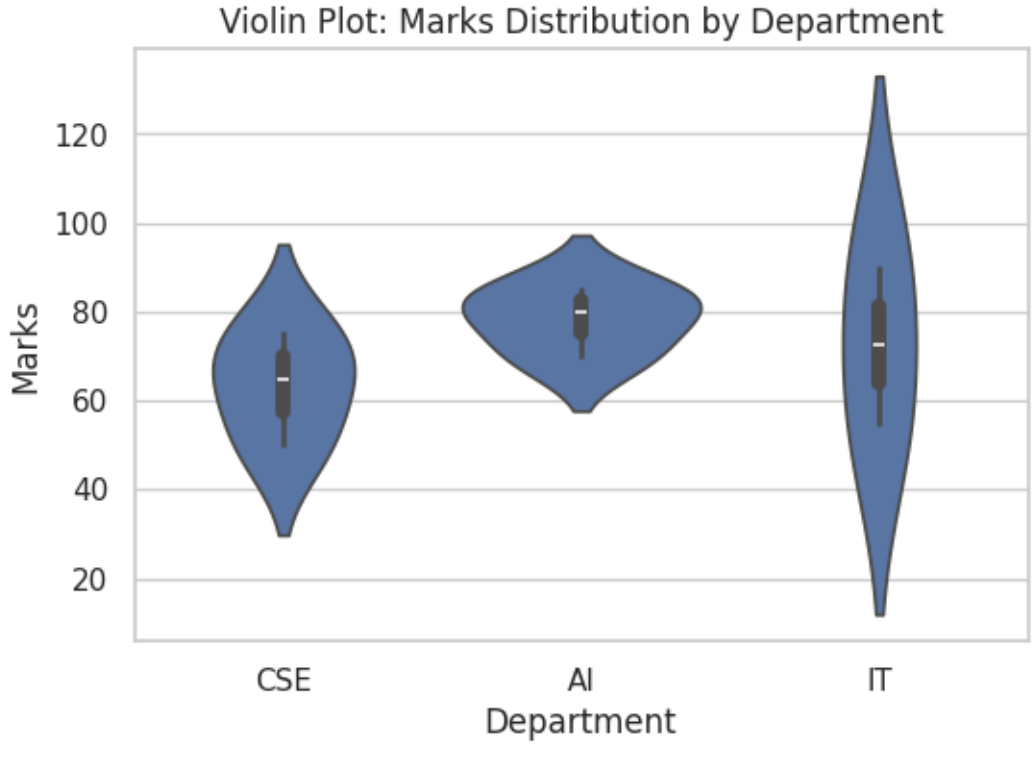
```
Sample DataFrame:
  Student Department  Gender  Study_Hours  Marks  Attendance
0    Arun        CSE    Male            2     50          70
1    Bala         AI    Male            4     70          85
2  Chitra        CSE  Female            3     65          80
3   Divya         AI  Female            5     85          95
4   Ezhil         IT    Male            6     90          90
5  Farhan         IT    Male            2     55          65
6  Geetha        CSE  Female            4     75          88
7    Hari         AI    Male            5     80          92
```

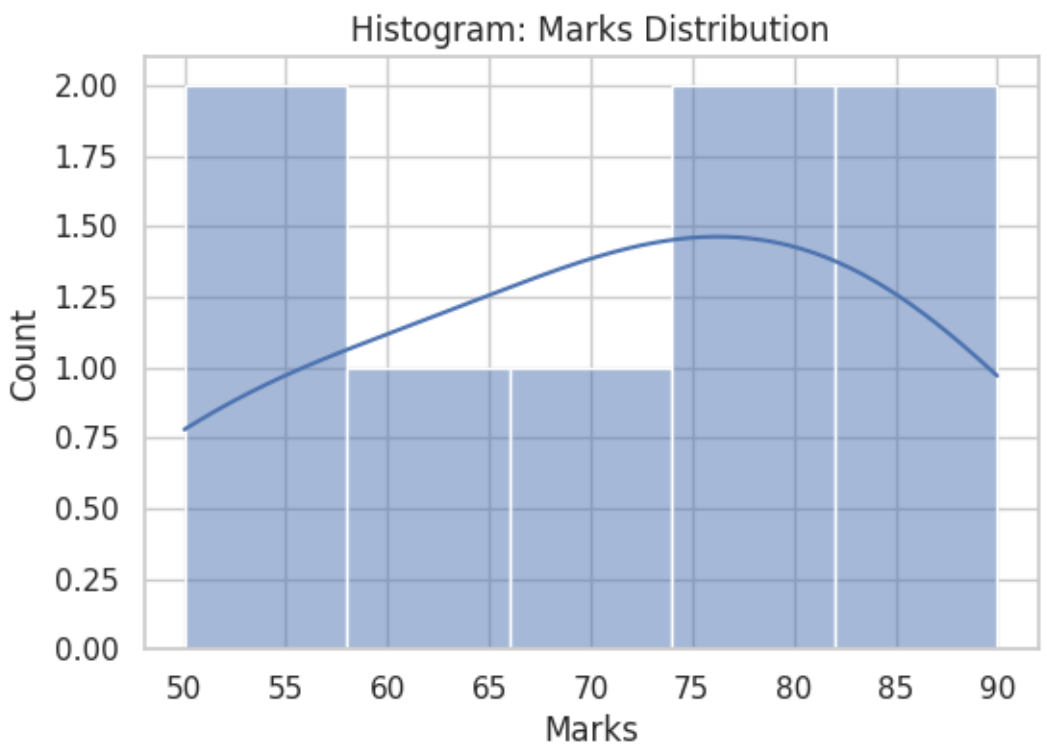

Line Plot: Study Hours vs Marks
Marks
90
85
80
75
70
65
60
55
50
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
Study_Hours
Scatter Plot: Study Hours vs Marks
Department
CSE
AI
IT
Marks
Study_Hours

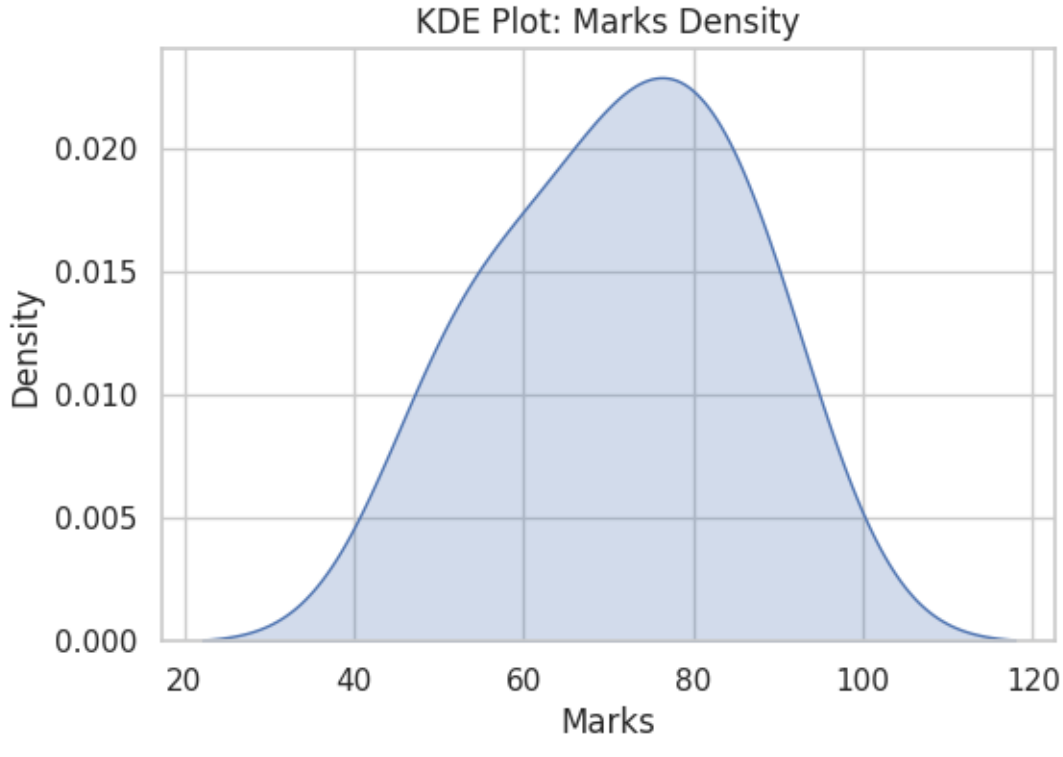

Bar Plot: Average Marks by Department
Marks
0
20
40
60
80
CSE
AI
IT
Department

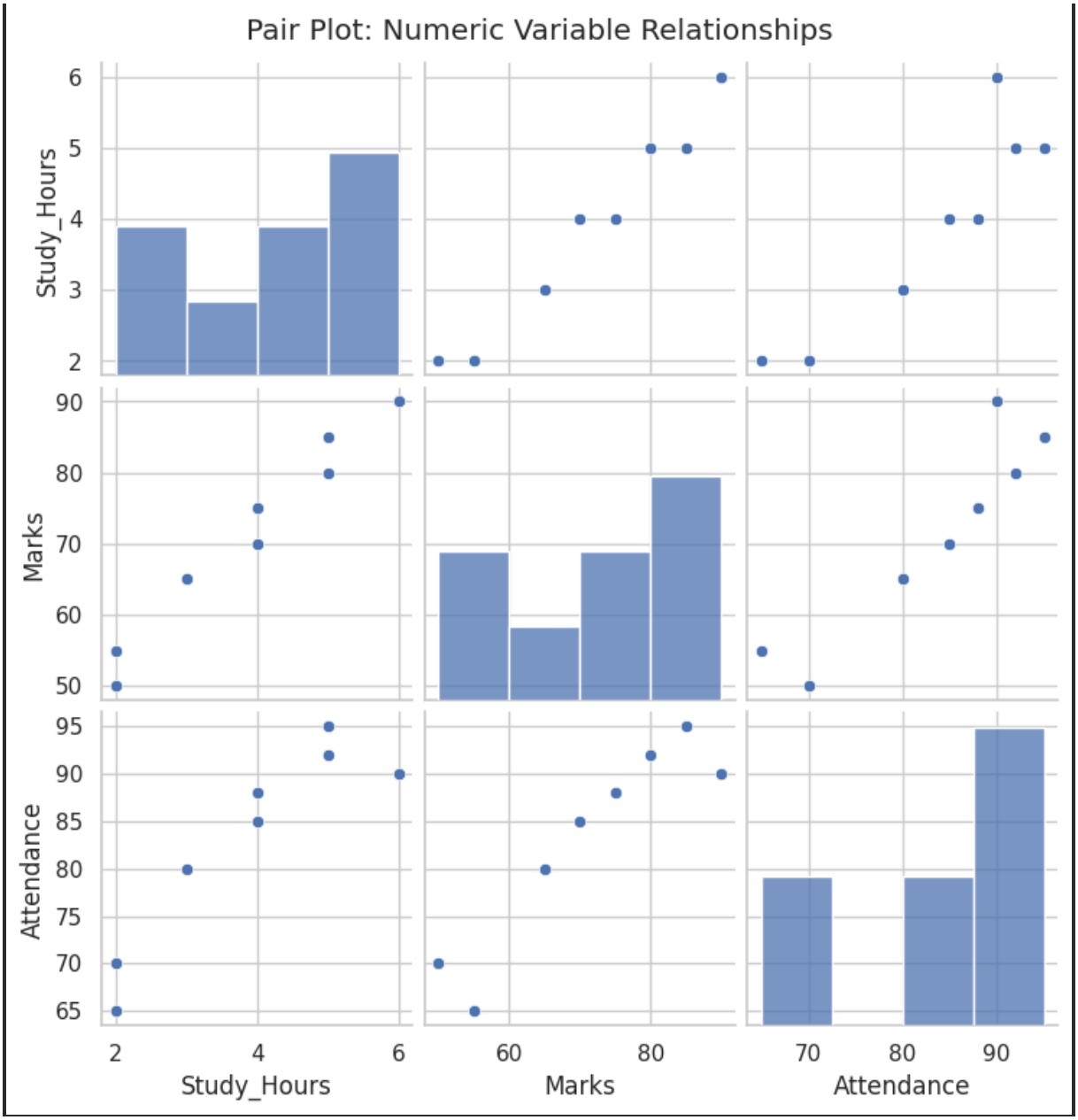

Count Plot: Number of Students by Department
count
0.0
0.5
1.0
1.5
2.0
2.5
3.0
CSE
AI
IT
Department

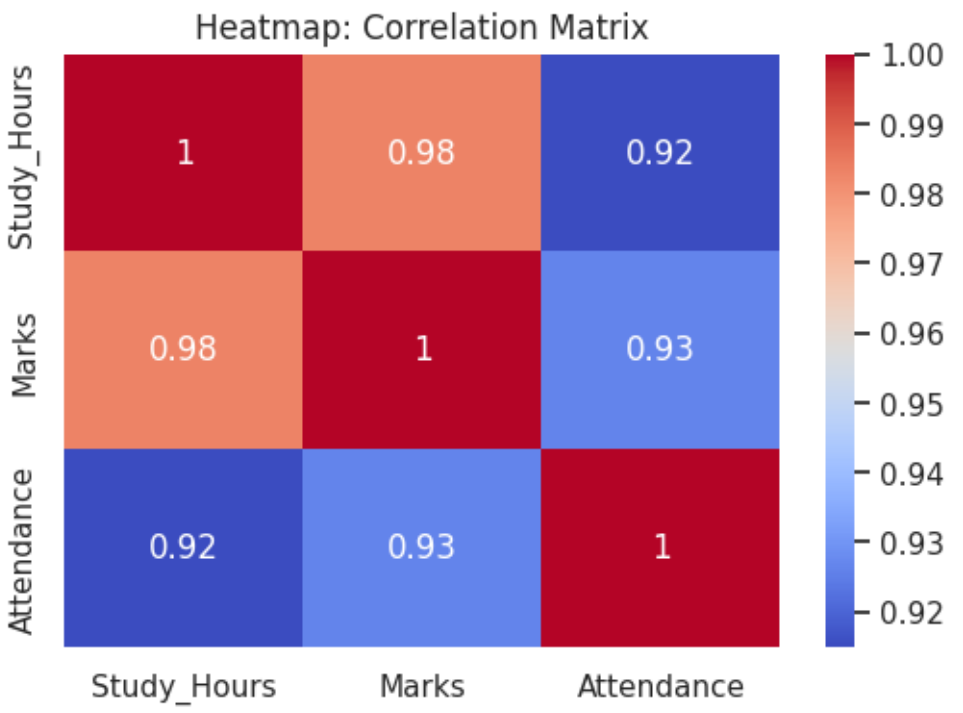

Box Plot: Marks Distribution by Department
Marks
90
85
80
75
70
65
60
55
50
CSE
AI
IT
Department


Violin Plot: Marks Distribution by Department
Marks
120
100
80
60
40
20
CSE
AI
IT
Department

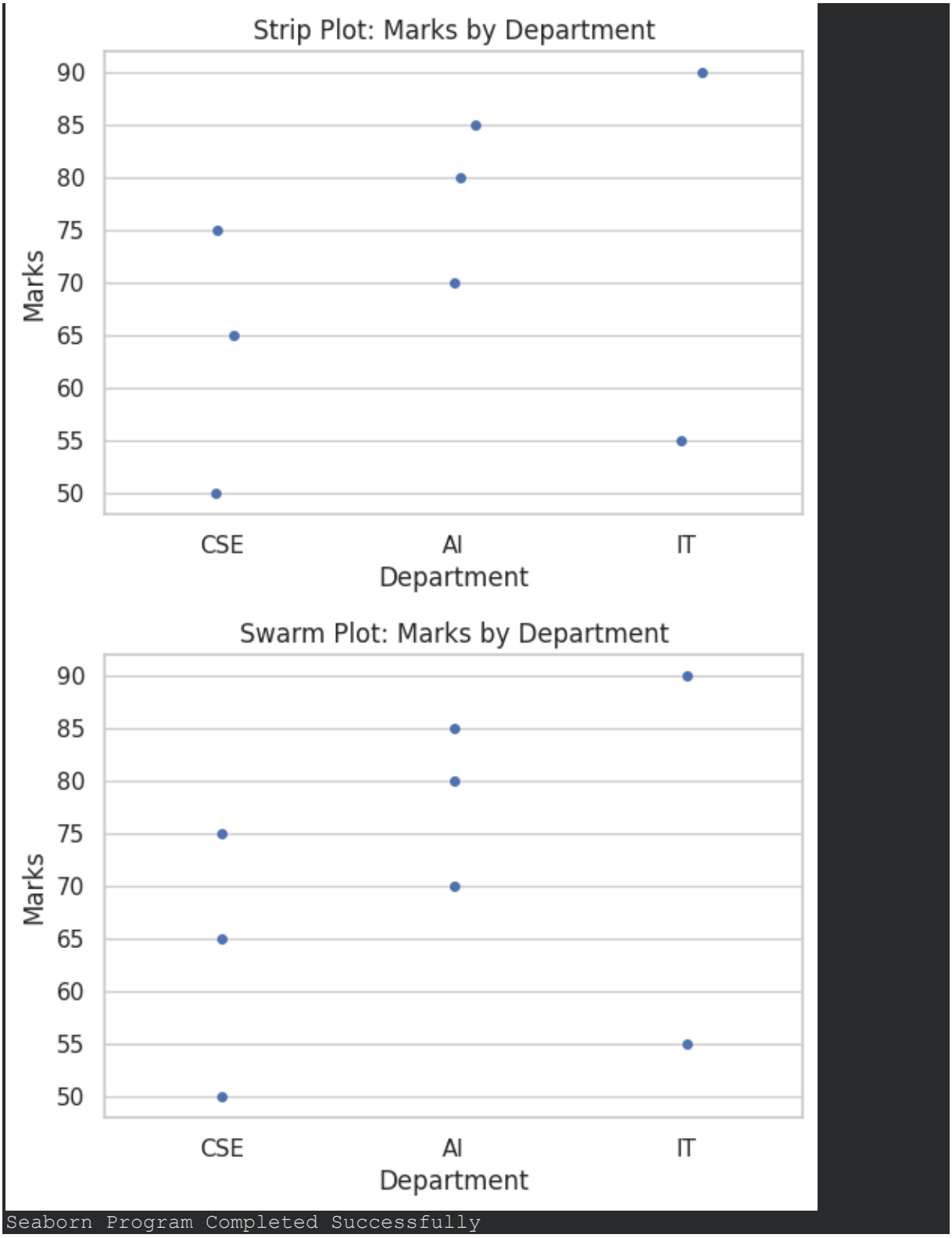

Histogram: Marks Distribution
Count
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
50
55
60
65
70
75
80
85
90
Marks


KDE Plot: Marks Density
Density
0.020
0.015
0.010
0.005
0.000
20
40
60
80
100
120
Marks


Pair Plot: Numeric Variable Relationships
Study_Hours
Marks
Attendance
2
4
6
60
80
70
80
90
50
65
75
85
95


Heatmap: Correlation Matrix
Study_Hours
Marks
Attendance
1
0.98
0.92
0.98
1
0.93
0.92
0.93
1
Study_Hours
Marks
Attendance
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0.92


Regression Plot: Study Hours vs Marks
Marks
90
80
70
60
50
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
Study_Hours




```
Seaborn Program Completed Successfully
```